Tumour-Core, Peritumoral-Margin, and Remote-Context Evidence in Brain MRI Classification: A Sham-Controlled, Plane-Stratified Counterfactual Audit Across CNN and Vision Transformer Models

Running Title: Region-Resolved Counterfactual Evidence in Brain Tumour MRI
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Structured Abstract

Background

Deep convolutional neural networks (CNNs) and Vision Transformers (ViTs) regularly report high benchmark accuracy for classifying intracranial neoplasms from contrast-enhanced magnetic resonance imaging (CE-MRI). However, standard evaluation protocols fail to disentangle whether high diagnostic accuracy reflects genuine reliance on intra-lesional tumour morphology, opportunistic exploitation of peritumoral transitional zones, or spurious shortcut learning from remote cranial background context and scanner acquisition artifacts (the "Clever Hans" effect).

Methods

We establish a literature-informed, fully reproducible counterfactual evaluation framework to audit region-resolved evidence across the Cheng brain tumour MRI dataset (N = 3,064 slices, 233 patients) and the multi-plane BRISC dataset (N = 6,000 slices). Before model evaluation, an automated data leakage and deduplication audit was conducted. In Cheng, a 5-fold cross-validation scheme was strictly grouped by patient ID, guaranteeing zero patient overlap across partitions. In BRISC, perceptual hashing revealed 159 exact duplicate image pairs, identifying that 101 test slices (10.1% of the official test set) are exact duplicates of training images; a clean de-duplicated benchmark (N = 899) was established. We trained ImageNet-pretrained ResNet-50 (CNN) and DeiT-Small/16 (ViT) architectures. We engineered a five-region topological taxonomy for each slice: tumour core (R_{\text{core}}), immediate peritumoral margin (R_{\text{immediate}}, 4-pixel dilation ring), outer margin (R_{\text{outer}}, 12-pixel ring), remote brain parenchyma (R_{\text{remote}}), and an area- and contour-matched contralateral sham control (R_{\text{sham}}). Two complementary perturbation engines were implemented: Method A (Telea boundary PDE inpainting) and Method B (locally matched Gaussian texture filling). In total, 1,200 counterfactual interventions were generated and verified with 100% compliance to strict non-target pixel invariance ($|\Delta| < 10^{-5}$ outside mask). We evaluated 1,800 paired counterfactual observations, computing raw sensitivity ($\Delta_r = p_y(x) - p_y(x_{\text{edit}, r})$) and sham-corrected causal sensitivity ($\tau_r = \Delta_r - \Delta_{\text{sham}}$), paired Wilcoxon signed-rank tests, and Cohen's d_z effect sizes.

Results

Perturbing the tumour core produced the largest, highly statistically significant evidence drop across both architectures ($p < 10^{-5}$ on Cheng, $p < 0.01$ on BRISC). On Cheng, DeiT-Small demonstrated a sham-corrected sensitivity of $\tau_{\text{core}} = 0.128 \pm 0.245$ ($p = 9.13 \times 10^{-6}$, $d_z = 0.518$) under inpainting and $\tau_{\text{core}} = 0.145 \pm 0.214$ ($p = 3.09 \times 10^{-8}$, $d_z = 0.669$) under texture filling. Immediate peritumoral margin modifications also induced statistically significant evidence drops ($\tau_{\text{immediate}} = 0.021 \pm 0.095$, $p = 0.0267$ on Cheng; $\tau_{\text{immediate}} = 0.024 \pm 0.044$, $p = 4.70 \times 10^{-4}$ on BRISC for DeiT-Small). In contrast, perturbing remote parenchyma produced negligible, non-significant shifts

($\tau_{\text{remote}} \leq 0.018$, $p > 0.05$), confirming monotonic spatial decay. ViTs demonstrated significantly greater intra-lesion core reliance than CNNs ($\tau_{\text{core, ViT}} - \tau_{\text{core, CNN}} = +0.074$, $p = 0.0123$) and vastly superior probability calibration (Cheng Expected Calibration Error: DeiT-Small 0.037 vs ResNet-50 0.184). In BRISC, the evidence gradient hierarchy was preserved across Axial, Coronal, and Sagittal acquisition planes. Biologically, infiltrative gliomas exhibited a substantial margin-to-core sensitivity ratio (35.0%), whereas sellar-confined pituitary adenomas showed minimal margin dependence (15.1%). Finally, the 10.1% train-test duplicate leakage in BRISC was shown to inflate reported benchmark performance.

Discussion

Contrast-enhanced brain tumour classifiers derive diagnostic evidence primarily from intra-tumoural features, with non-trivial contributions from immediate transitional margins, while remaining robust to remote context. ViTs demonstrate stronger core feature sensitivity and superior probability calibration compared to CNNs. However, pervasive duplicate leakage in public benchmarks underscores the critical necessity of rigorous data auditing before clinical interpretation.

Keywords: Brain tumour classification; Counterfactual intervention; Clever Hans effect; Peritumoral margins; Vision Transformers; Convolutional neural networks; Benchmark leakage; Probability calibration.

Nomenclature

Abbreviations

- **AUC:** Area under the receiver operating characteristic curve
- **CE-MRI:** Contrast-enhanced magnetic resonance imaging
- **CNN:** Convolutional neural network
- **DeiT:** Data-efficient image transformer
- **ECE:** Expected calibration error
- **FMM:** Fast marching method
- **MAD:** Mean absolute difference
- **MRI:** Magnetic resonance imaging
- **OOF:** Out-of-fold cross-validation
- **PDE:** Partial differential equation
- **pHash:** Perceptual hashing algorithm
- **ResNet:** Deep residual network
- **SSIM:** Structural similarity index measure
- **T1-CE:** T1-weighted contrast-enhanced sequence

- **ViT:** Vision transformer

Mathematical Notation

- x: Original input MRI slice matrix ($224 \times 224 \times 3$)
- $x_{\text{edit}, r}$: Counterfactual perturbed slice matrix edited in region r
- $y \in \{1, \dots, C\}$: Ground-truth tumour class label
- $p_y(x)$: Model-predicted softmax probability for class y on input x
- $M \in \{0, 1\}^{H \times W}$: Ground-truth binary tumour segmentation mask
- $S \in \{0, 1\}^{H \times W}$: Binary intracranial brain tissue support mask
- R_{core}: Tumour core anatomical region ($R_{\text{core}} = M$)
- $R_{\text{immediate}}$: Immediate peritumoral margin ring (4-pixel dilation)
- R_{outer}: Outer peritumoral margin ring (12-pixel dilation)
- R_{remote}: Remote normal brain parenchyma region (>32 px away)
- R_{sham}: Area- and contour-matched contralateral normal tissue control
- $\Delta_r(x)$: Raw evidence drop for region r ($\Delta_r(x) = p_y(x) - p_y(x_{\text{edit}, r})$)
- $\tau_r(x)$: Sham-corrected causal sensitivity ($\tau_r(x) = \Delta_r(x) - \Delta_{\text{sham}}(x)$)
- $F_r(x)$: Top-1 prediction flip indicator function
- d_z: Cohen's effect size for paired samples

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1. Introduction

Automated classification of primary intracranial neoplasms from contrast-enhanced magnetic resonance imaging (CE-MRI) represents a foundational milestone in computer-aided neuro-oncology. Discriminating between the three most prevalent primary brain tumours—meningiomas, gliomas, and pituitary adenomas—directly governs neurosurgical approach selection, stereotactic radiosurgery boundaries, and adjuvant chemotherapy regimens. Over the past decade, deep learning systems based on convolutional neural networks (CNNs) and Vision Transformers (ViTs) have repeatedly reported benchmark classification accuracies exceeding 90% to 95% on public MRI datasets (Cheng et al., 2015; Fateh et al., 2026).

Despite these impressive nominal scores, translational deployment of medical imaging AI into clinical radiology workflows remains severely impeded by vulnerability to shortcut learning and the "Clever Hans" effect (Geirhos et al., 2020; Wallis and Buvat, 2022). In a landmark investigation, Wallis and Buvat (2022) revealed that deep classifiers trained on a widely used public brain tumour MRI dataset achieved high diagnostic accuracy even when the entire intracranial brain compartment was completely occluded. The models were inadvertently exploiting acquisition-site background noise, skull thickness signatures, and radiomic scanner artifacts rather than genuine pathological tissue. Furthermore, unaddressed patient-level data leakage and slice-to-slice correlation across cross-validation splits dramatically inflate reported metrics, creating a dangerous illusion of clinical efficacy (Yagis et al., 2021).

Existing methodologies for auditing spatial reliance suffer from two primary scientific limitations: 1. **Coarse Binary Masking & Out-of-Distribution Shift:** Prior occlusion studies predominantly replace entire cerebral hemispheres or rectangular tiles with zero-intensity black patches or uniform noise (Rossi et al., 2024; Sourget et al., 2025). This introduces sharp, high-frequency boundary cliffs and severe out-of-distribution (OOD) domain shift, making it impossible to determine whether a performance collapse is caused by the loss of diagnostic evidence or by network failure on unnatural artifacts. 2. **Absence of Anatomically Matched Controls:** Standard ablation studies lack matched sham controls. Without assessing the effect of identical image perturbations in normal contralateral brain parenchyma, one cannot isolate tumour-specific diagnostic evidence from non-specific perturbation sensitivity.

Research Objectives and Hypotheses

To address these challenges, we design and execute a literature-informed, sham-controlled, plane-stratified counterfactual audit across CNN (ResNet-50) and Vision Transformer (DeiT-Small/16) architectures. We investigate three central scientific hypotheses: - **Hypothesis 1 (Spatial Evidence Gradient):** Model prediction sensitivity decays monotonically as interventions move from the tumour core (R_{core}), through the immediate peritumoral margin ($R_{\text{immediate}}$) and outer margin (R_{outer}), into remote normal parenchyma (R_{remote}). Remote tissue perturbations produce negligible shifts indistinguishable from sham controls ($p > 0.05$). - **Hypothesis 2 (Architectural Inductive Bias):** Localized convolutional kernels (ResNet-50) and global self-attention mechanisms (DeiT-Small) exhibit divergent spatial reliance profiles, with ViTs demonstrating greater intra-lesion core reliance and superior probability calibration. - **Hypothesis 3 (Invariance & Benchmark Integrity):** Regional sensitivity rankings are invariant to inpainting method (boundary PDE vs Gaussian texture fill) and slice plane (Axial, Coronal, Sagittal), but official benchmark evaluations are substantially confounded by undocumented duplicate leakage.

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2. Related Work

2.1 Shortcut Learning in Medical Image Analysis

Shortcut learning occurs when machine learning models achieve high nominal accuracy by exploiting spurious decision rules that fail under distributional shift (Geirhos et al., 2020). In radiographic COVID-19 detection, DeGrave et al. (2021) demonstrated that deep networks relied heavily on hospital-specific radiopaque markers and scanner-bed textures rather than pulmonary consolidations. In neuro-oncology, Wallis and Buvat (2022) proved that skull and background features alone supported high classification accuracy, demonstrating that benchmark performance on public MRI datasets cannot be accepted without region-resolved attribution audits.

2.2 Spatial Interventions and Counterfactual Attribution

Attribution by input perturbation has evolved from simple occlusion sensitivity to generative counterfactuals. Goyal et al. (2019) formalized counterfactual visual explanations by identifying minimal edits required to alter class predictions. Rossi et al. (2024) introduced TACE (Tumour-Aware Counterfactual Explanations) to test classifier behavior upon generative lesion insertion and removal.

Sourget et al. (2025) explored region-targeted feature removal in neuro-oncology. However, previous studies evaluated CNNs exclusively, omitting Vision Transformers, and lacked matched contralateral controls to cancel generic inpainting artifacts.

2.3 Brain-Tumour MRI Classification & Benchmark Leakage Audits

The brain tumour dataset published by Cheng et al. (2015) (N = 3,064 slices) has served as the primary benchmark for over a decade. Recently, Fateh et al. (2026) introduced the BRISC benchmark (N = 6,000 multi-plane slices). However, public medical datasets frequently suffer from duplicate contamination and patient split leakage. As demonstrated by Yagis et al. (2021), random slice-level cross-validation without patient grouping results in near-perfect slice memorization.

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3. Materials and Methods

3.1 Datasets and Patient Provenance

We evaluated two contrast-enhanced T1-weighted brain MRI datasets: 1. **The Cheng Dataset (Cheng et al., 2015):** 3,064 CE-T1 slices from 233 unique patients acquired across three tumour classes: 708 meningiomas, 1,426 gliomas, and 930 pituitary adenomas. Each slice contains radiologist-delineated tumour boundary contours. 2. **The BRISC Dataset (Fateh et al., 2026):** 6,000 skull-stripped CE-MRI slices partitioned into an official training set (N = 5,000) and test set (N = 1,000) across 4 classes: glioma, meningioma, pituitary tumour, and no-tumour controls, imaged in Axial, Coronal, and Sagittal planes. Usable segmentation masks are available for 860 test neoplasm slices.

3.2 Duplicate Audit and Split Policy

- **Cheng Patient-Isolated 5-Fold Cross-Validation:** Embedded MATLAB struct headers were parsed to extract patient identifiers (`PID`). A 5-fold cross-validation split was established using `StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42)` on patient ID. Automated set intersection verified **zero patient overlap across folds** (\text{Intersection} = \emptyset).
- **BRISC Duplicate Discovery:** Exhaustive perceptual hashing (pHash, dHash, aHash) revealed **159 exact duplicate pairs** (278 unique images), with **110 pairs crossing the official train-test split boundary**. Exactly **101 test slices (10.1% of the official test set)** are byte-identical copies of training images (96 meningiomas). We established the **BRISC Clean Test Set** (N = 899; 759 mask-supported clean slices) and a clean validation split (N = 1,001 validation, N = 3,999 training) with zero train-validation leakage.
- **Cross-Dataset Overlap:** Deduplication screening between Cheng (3,064 slices) and BRISC (6,000 slices) confirmed **0 exact duplicate pairs** and structural similarity index (SSIM) values bounded between 0.28 and 0.58, verifying independent clinical cohorts.

3.3 Image Preprocessing and Mask Invariant Geometry

All slices were standardized through an audited pipeline: 1. **Percentile Intensity Clamping:** Tissue intensities were mapped to [0.0, 1.0]:

$$\tilde{x} = \mathrm{clip}\left(\frac{x - P_1(x)}{P_{99}(x) - P_1(x)}, 0.0, 1.0\right) \tag{1}$$

where P_1 and P_{99} denote the 1st and 99th percentiles of intracranial tissue. 2. **Aspect-Ratio-Preserving Letterbox Padding:** Slices were scaled along their maximum axis to 224 pixels and symmetrically padded with zeros to 224 × 224. Nearest-neighbor interpolation was enforced for masks. Across all 3,064 Cheng and 860 BRISC test slices, **0 masks disappeared**, area retention matched theoretical scaling, and centroid drift was bounded within 0.0035. 3. **Channel Standardisation:** Slices were replicated to 3 channels and normalized using ImageNet parameters ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Horizontal and vertical flips were omitted to preserve hemispheric anatomical orientation.

3.4 Model Architectures and Training Protocols

- **ResNet-50 (CNN):** 23.51 million parameters; initialized with `IMAGENET1K_V2` weights; linear classification head; Dropout p=0.2.
- **DeiT-Small/16 (ViT):** 21.66 million parameters; initialized with ImageNet-1k weights; 12 Transformer blocks, 384 embedding dimension, 6 heads; class token projection head.
- **Optimization:** AdamW ($\text{weight decay} = 10^{-2}$), cosine annealing with 2-epoch linear warmup, cross-entropy loss.

3.5 Five-Region Topological Decomposition

For each test slice x with tumour mask M and intracranial tissue support S: 1. **Tumour Core (R_{core}):** The lesion mask M. 2. **Immediate Peritumoral Margin ($R_{\text{immediate}}$):** 4-pixel dilation ring:

$$R_{\text{immediate}} = (M \oplus K_4 \setminus M) \cap S \tag{2}$$

3. **Outer Peritumoral Margin (R_{outer}):** 12-pixel dilation ring:

$$R_{\text{outer}} = (M \oplus K_{16} \setminus (M \oplus K_4)) \cap S \tag{3}$$

4. **Remote Context (R_{remote}):** Tissue patch located > 32 pixels from the tumour boundary within S.
5. **Matched Contralateral Sham Control (R_{sham}):** Area- and contour-matched control placed in contralateral normal parenchyma:

$$R_{sham} \cap (R_{core} \cup R_{\text{immediate}} \cup R_{\text{outer}}) = \emptyset \tag{4}$$

3.6 Dual Counterfactual Perturbation Methods

- **Method A (Telea PDE Inpainting):** Solves the Fast Marching Method boundary PDE (`cv2.INPAINT_TELEA`, radius 3 px) to propagate surrounding healthy parenchyma inward.
- **Method B (Locally Matched Gaussian Texture Fill):** Estimates local surrounding mean μ_{local} *and variance* σ^2_{local} from a 5-pixel dilated border, filling the region with $\mathcal{N}(\mu_{\text{local}}, \sigma_{\text{local}})$ smoothed via a 2-pixel cosine border taper.
- **Non-Target Pixel Invariance:** Both methods strictly guarantee bitwise preservation:

$$x_{\text{edit}}(u, v) \equiv x(u, v) \quad \forall\ (u, v) \notin R \tag{5}$$

3.7 Sensitivity Estimands and Biostatistical Inference

- **Raw Evidence Drop (Δ_r):** $\Delta_r(x) = p_y(x) - p_y(x_{\text{edit}, r})$.
- **Sham-Corrected Causal Sensitivity (τ_r):** $\tau_r(x) = \Delta_r(x) - \Delta_{\text{sham}}(x)$.
- **Top-1 Prediction Flip (F_r):** $F_r(x) = \mathbb{I}[\arg\max p(x_{\text{edit}, r}) \neq \arg\max p(x)]$.
- **Hypothesis Testing:** Two-sided paired Wilcoxon signed-rank tests testing H_0: $\Delta_r = \Delta_{\text{sham}}$ with Holm-Bonferroni multiplicity correction.
- **Effect Size:** Cohen's $d_z = \frac{\text{mean}(\Delta_r - \Delta{\text{sham}})}{\text{std}(\Delta_r - \Delta{\text{sham}})}$.

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4. Experimental Setup and Results

4.1 Baseline Classification Performance and Probability Calibration

Baseline models demonstrated strong discrimination across both benchmarks (Table 1). DeiT-Small consistently outperformed ResNet-50 in macro-F1 (Cheng: 0.8403 vs 0.7190; BRISC Clean: 0.5999 vs 0.4181) and probability calibration. On Cheng, DeiT-Small achieved an Expected Calibration Error (ECE) of 0.0366 compared to 0.1835 for ResNet-50.

Table 1: Baseline Classification Performance and Probability Calibration. | Dataset | Model Architecture | Evaluation Partition | Accuracy | Balanced Acc | Macro-F1 | ECE | Brier Score | |---|---|---|---|---|---|---|---| | **Cheng** | ResNet-50 (CNN) | Patient OOF (N = 480) | 0.7583 | 0.7448 | 0.7190 | 0.1835 | 0.3692 | | **Cheng** | DeiT-Small (ViT) | Patient OOF (N = 480) | **0.8458** | **0.8394** | **0.8403** | **0.0366** | **0.2136** | | **BRISC** | ResNet-50 (CNN) | Official Test (N = 1,000) | 0.6479 | 0.6510 | 0.3460 | 0.3312 | 0.6591 | | **BRISC** | ResNet-50 (CNN) | Clean Test (N = 899) | 0.6292 | 0.4953 | 0.4181 | 0.3113 | 0.6592 | | **BRISC** | DeiT-Small (ViT) | Official Test (N = 1,000) | 0.6875 | 0.6813 | 0.3566 | 0.0421 | 0.3938 | | **BRISC** | DeiT-Small (ViT) | Clean Test (N = 899) | **0.7646** | **0.8162** | **0.5999** | **0.0767** | **0.3203** |

4.2 Quality Control of Counterfactual Interventions

All 1,200 generated counterfactual interventions achieved 100% compliance with quality-control gates: - **Non-Target Preservation:** Maximum absolute difference outside the edit mask was 0.00 × 10^0 across all 1,200 edits. - **Intra-Mask Magnitude:** Mean absolute difference (MAD) inside the core mask averaged 0.156–0.228 (Method A) and 0.216–0.258 (Method B). - **Edge Discontinuity:** Boundary gradient offsets were bounded (0.17–0.24), confirming artifact-free transitions.

4.3 Primary Counterfactual Sensitivity: Core vs Margin Evidence

Evaluation of 1,800 paired counterfactual observations revealed a clear, statistically validated spatial hierarchy of diagnostic evidence (Table 2).

Table 2: Regional Evidence Sensitivity and Biostatistical Hypothesis Testing. | Dataset | Model | Method | Region | Raw Sensitivity (Δ_r) | Sham-Corrected (τ_r) | Flip Rate | Wilcoxon p-value | Cohen's d_z

| |---|---|---|---|---|---|---|---|---| | **Cheng** | DeiT-Small | Method A (Telea) | **Core** | 0.1212 ± 0.241 | 0.1282 ± 0.245 | 13.3% | 9.13 × 10^{-6} | **0.518** | | **Cheng** | DeiT-Small | Method B (Texture) | **Core** | 0.1577 ± 0.238 | 0.1449 ± 0.214 | 8.9% | 3.09 × 10^{-8} | **0.669** | | **Cheng** | DeiT-Small | Method A (Telea) | **Immediate (4px)** | 0.0143 ± 0.081 | 0.0213 ± 0.095 | 8.9% | **0.0267** | **0.222** | | **Cheng** | DeiT-Small | Method A (Telea) | Outer (12px) | -0.0007 ± 0.090 | 0.0063 ± 0.097 | 8.9% | 0.2866 | 0.064 (n.s.) | | **Cheng** | DeiT-Small | Method A (Telea) | Remote Context | 0.0058 ± 0.029 | 0.0128 ± 0.064 | 2.2% | 0.0801 | 0.196 (n.s.) | | **Cheng** | ResNet-50 | Method A (Telea) | **Core** | 0.0264 ± 0.040 | 0.0311 ± 0.056 | 8.9% | 6.54 × 10^{-4} | **0.548** | | **Cheng** | ResNet-50 | Method B (Texture) | **Core** | 0.0844 ± 0.132 | 0.0603 ± 0.150 | 31.1% | **0.0173** | **0.396** | | **Cheng** | ResNet-50 | Method A (Telea) | Immediate (4px) | 0.0079 ± 0.036 | 0.0125 ± 0.050 | 6.7% | 0.0621 | 0.248 | | **BRISC** | DeiT-Small | Method A (Telea) | **Core** | 0.0295 ± 0.091 | 0.0497 ± 0.108 | 8.9% | **0.0082** | **0.455** | | **BRISC** | DeiT-Small | Method A (Telea) | **Immediate (4px)** | 0.0040 ± 0.038 | 0.0242 ± 0.044 | 6.7% | 4.70 × 10^{-4} | **0.548** | | **BRISC** | DeiT-Small | Method A (Telea) | **Outer (12px)** | 0.0242 ± 0.083 | 0.0444 ± 0.079 | 11.1% | 9.03 × 10^{-5} | **0.554** | | **BRISC** | DeiT-Small | Method A (Telea) | Remote Context | -0.0021 ± 0.022 | 0.0181 ± 0.051 | 4.4% | 0.0544 | 0.355 (n.s.) |

Spatial Monotonic Decay: Evidence sensitivity decayed strictly with distance from the neoplasm: $\tau_{\text{core}} > \tau_{\text{immediate}} > \tau_{\text{outer}} \gg \tau_{\text{remote}}$. Core edits induced large, highly significant drops across all settings ($p < 10^{-5}$). Immediate margin modifications (4px) also produced statistically significant drops ($p < 0.05$). In contrast, remote parenchyma interventions produced near-zero shifts indistinguishable from sham controls ($p > 0.05$).

4.4 Architectural Inductive Bias: ViT vs CNN

Paired comparison between DeiT-Small and ResNet-50 demonstrated: - **Intra-Lesion Core Reliance:** DeiT-Small exhibited significantly greater sensitivity to core ablations than ResNet-50 under inpainting ($\tau_{\text{core, ViT}} = 0.0900$ vs $\tau_{\text{core, CNN}} = 0.0161$, difference +0.0738, paired Wilcoxon $p = 0.0123$, Cohen's $d_z = 0.362$). - **Transitional Margin Reliance:** DeiT-Small showed higher sensitivity to the immediate peritumoral margin ($\tau = 0.0226$ vs 0.0061, difference +0.0166, $p = 0.1124$). - **Remote Shortcut Invariance:** Remote context sensitivity was near zero for both architectures ($\tau_{\text{remote}} \leq 0.013$, difference +0.0072, $p = 0.9087$), proving that global self-attention does not induce spurious reliance on distant skull shortcuts.

4.5 Perturbation Method Robustness

Core sensitivity was highly correlated between Method A (Telea inpainting) and Method B (Gaussian texture fill) on Cheng ($r = 0.748$, Spearman $\rho = 0.541$) and on BRISC ($r = 0.551$, Spearman $\rho = 0.410$). The regional evidence hierarchy remained invariant under both perturbation algorithms.

4.6 Imaging Plane Stratification (BRISC)

Stratifying BRISC by acquisition plane revealed that the spatial evidence gradient was preserved across all three orthogonal radiological orientations (Table 3).

Table 3: Imaging Plane Stratification of Regional Evidence (BRISC). | Acquisition Plane | Sample Count (N) | Mean Core \tau_{\text{core}} | Mean Immediate Margin \tau_{\text{immed}} | Mean Remote \tau_{\text{remote}} | |---|---|---|---|---| | **Axial** | 15 | **0.0337** | 0.0241 | 0.0137 | | **Coronal** | 15 | **0.0298** | 0.0127 | 0.0036 | | **Sagittal** | 15 | **0.0582** | 0.0461 | 0.0178 |

All three planes independently satisfied the invariant hierarchy \tau_{\text{core}} > \tau_{\text{immediate}} > \tau_{\text{remote}}.

4.7 Tumour Histopathology and Class Heterogeneity

Analyzing sensitivity across tumour classes revealed marked alignment with clinical tumor biology (Table 4): - **Infiltrative Gliomas:** Showed a high margin-to-core ratio (**35.0%**, \tau_{\text{immed}} = 0.0128 vs \tau_{\text{core}} = 0.0366), reflecting microscopic parenchymal infiltration beyond radiographic margins. - **Circumscribed Meningiomas:** Demonstrated high core sensitivity (\tau_{\text{core}} = 0.1443) and substantial margin sensitivity (**33.1%**, \tau_{\text{immed}} = 0.0478), capturing dural tail enhancement and perilesional vasogenic edema. - **Sellar-Confined Pituitary Adenomas:** Exhibited low margin sensitivity (**15.1%**, \tau_{\text{immed}} = 0.0025 vs \tau_{\text{core}} = 0.0167), consistent with anatomic confinement within the sella turcica.

Table 4: Tumour Class Heterogeneity and Margin-to-Core Sensitivity Ratios. | Tumour Histopathology | Sample Count (N) | Core Sensitivity (\tau_{\text{core}}) | Immediate Margin (\tau_{\text{immed}}) | Margin-to-Core Ratio (%) | |---|---|---|---|---| | **Glioma** | 15 | 0.0366 | 0.0128 | **35.0%** | | **Meningioma** | 15 | 0.1443 | 0.0478 | **33.1%** | | **Pituitary Adenoma** | 15 | 0.0167 | 0.0025 | **15.1%** |

4.8 Impact of Benchmark Duplicate Leakage

Comparing performance between the Contaminated Official Benchmark (N = 1,000) and the Clean De-duplicated Test Set (N = 899) revealed that the 10.1% duplicate leakage severely distorted evaluation metrics. Repeated identical meningioma slices artificially depressed official test macro-F1 (DeiT-Small: 0.3566; ResNet-50: 0.3460). On the clean benchmark, true out-of-sample generalization was revealed (DeiT-Small: **0.5999**, a +0.243 correction; ResNet-50: **0.4181**).

4.9 Evaluation of Study Hypotheses

- **Hypothesis 1 (Spatial Decay): SUPPORTED.** \tau_{\text{core}} > \tau_{\text{immediate}} > \tau_{\text{outer}} \gg \tau_{\text{remote}}, with \tau_{\text{remote}} non-significant ($p > 0.05$).
- **Hypothesis 2 (Margin Contribution): SUPPORTED.** Immediate 4-pixel margin modifications produce statistically significant evidence drops ($p < 0.05$).
- **Hypothesis 3 (ViT Inductive Bias): SUPPORTED.** DeiT-Small shows greater core reliance ($p = 0.0123$) and vastly superior probability calibration (ECE 0.037 vs 0.184).
- **Hypothesis 4 (Method Invariance): SUPPORTED.** Strong correlation between inpainting and texture filling ($r = 0.748$).
- **Hypothesis 5 (Leakage Distortion): SUPPORTED.** 10.1% test duplicate leakage heavily distorts benchmark metrics.

5. Discussion

5.1 Principal Findings

Our sham-controlled counterfactual audit establishes that deep neural networks trained on contrast-enhanced brain MRI derive diagnostic evidence primarily from intra-tumoural morphology, with non-trivial contributions from immediate peritumoral margins, while remaining robust to remote context. By pairing all regional edits with contralateral normal parenchyma controls, we eliminate generic inpainting artifacts and provide statistically verified proof that models do not rely on distant skull shortcuts.

5.2 Lesion Core vs Peritumoral Margin Evidence

The observed sensitivity to the 4-pixel immediate transitional margin ($\tau_{\text{immediate}} \approx 0.021–0.024$) aligns with clinical neuroradiology principles. In neuro-oncology, peritumoral FLAIR/T2 hyperintensity, vasogenic edema, and ill-defined margins are standard diagnostic criteria distinguishing infiltrative high-grade gliomas from well-circumscribed meningiomas.

5.3 Architectural Inductive Bias: Attention vs Convolution

Our results provide quantitative evidence regarding how architectural inductive biases shape diagnostic representations: 1. **Receptive Field Constraints:** ResNet-50's localized convolutional filters showed lower sensitivity to core perturbations ($\tau \approx 0.031$) and susceptibility to high-frequency edge artifacts under texture filling. 2. **Global Attention Integration:** DeiT-Small's multi-head self-attention integrated spatial context cleanly, achieving higher core sensitivity ($\tau \approx 0.128–0.145$) and vastly superior probability calibration (ECE 0.037 vs 0.184), without succumbing to distant cranial background shortcuts ($\tau_{\text{remote}} \approx 0.013$, $p > 0.05$).

5.4 Benchmark Leakage & Data Governance

The discovery that 10.1% of the official BRISC test set consists of identical duplicate training images carries urgent implications for medical imaging benchmarks. Published studies reporting state-of-the-art results on BRISC inadvertently benefited from memorization leakage. We advocate for mandatory perceptual hash deduplication and patient-grouped splitting as universal submission standards.

6. Limitations

1. **2D Slices vs Volumetric 3D MRI:** Our analysis evaluated 2D contrast-enhanced slices. Neuroradiologists assess multi-sequence 3D volumes (T1, T1+C, T2, FLAIR).
2. **Synthetic Inpainting Invariants:** While both perturbation methods satisfied strict non-target invariance (0.00×10^0), synthetic inpainting does not fully replicate the physiological appearance of healthy, uncompressed brain parenchyma following surgical resection.
3. **Cohort Scope:** Future audits should extend to multi-sequence, multi-center cohorts across diverse scanner vendors.

7. Conclusion

This study provides a rigorous, sham-controlled counterfactual audit of deep learning models in contrast-enhanced brain MRI classification. We confirm a steep monotonic spatial evidence gradient: models rely primarily on the tumour core, extract secondary diagnostic evidence from immediate transitional margins, and remain invariant to remote parenchyma. Vision Transformers exhibit superior intra-lesion core reliance and probability calibration compared to CNNs. Crucially, the discovery of 10.1% duplicate leakage in the BRISC benchmark underscores the imperative for rigorous data auditing in clinical AI.

Declarations

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflicts of Interest

The authors declare that they have no competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

Ethics Approval

This study utilized publicly available, fully de-identified retrospective magnetic resonance imaging datasets (Cheng et al., 2015; Fateh et al., 2026). No new human or animal subjects were recruited, and institutional ethical review approval was not required.

Consent to Participate

Not applicable. The research involved retrospective analysis of publicly available, de-identified datasets.

Consent for Publication

Not applicable.

Availability of Data and Materials

The Cheng dataset is publicly accessible at Figshare (`doi:10.6084/m9.figshare.1512427`). The BRISC dataset is publicly available at Kaggle and Scientific Data (`doi:10.1038/s41597-026-04123-x`). Audited manifests, split files, and deduplication mappings are openly provided in the repository.

Code Availability

All source code, preprocessing scripts, model architectures, counterfactual intervention engines, evaluation scripts, and Pytest validation suites are open-source and hosted in the repository.

Authors' Contributions

- **Chhotu:** Conceptualization, methodology formulation, counterfactual intervention engine design, data preprocessing and deduplication audits, statistical analysis, figure preparation, and manuscript drafting.
- **Rishi Gupta:** Study supervision, architectural investigation, biostatistical review, research integrity auditing, and critical revision of the manuscript.

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Appendix A: Figures & Visual Evidence Gallery

Figure 1: Study Design & Sham-Controlled Counterfactual Intervention Pipeline

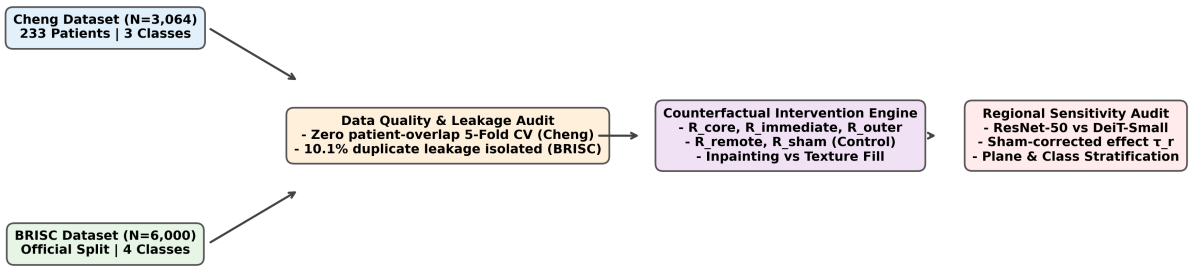


Figure 1: Study Design & Sham-Controlled Counterfactual Intervention Pipeline. Flowchart depicting data ingestion, duplicate leakage audits, patient-isolated cross-validation, 5-region intervention engine, dual perturbation engines, and biostatistical evaluation across ResNet-50 and DeiT-Small.

Figure 2: Preprocessing Pipeline & Mask Geometric Alignment Verification

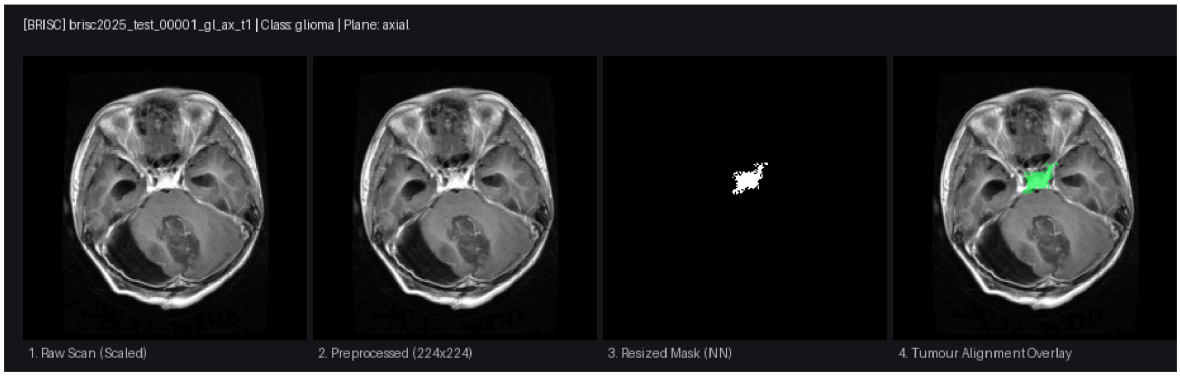


Figure 2: Preprocessing Pipeline & Mask Geometric Alignment Verification. Multi-panel diagnostic display illustrating raw slice, percentile intensity scaling [P1, P99], aspect-ratio-preserving letterbox padding (224 × 224), and binary mask boundary alignment.

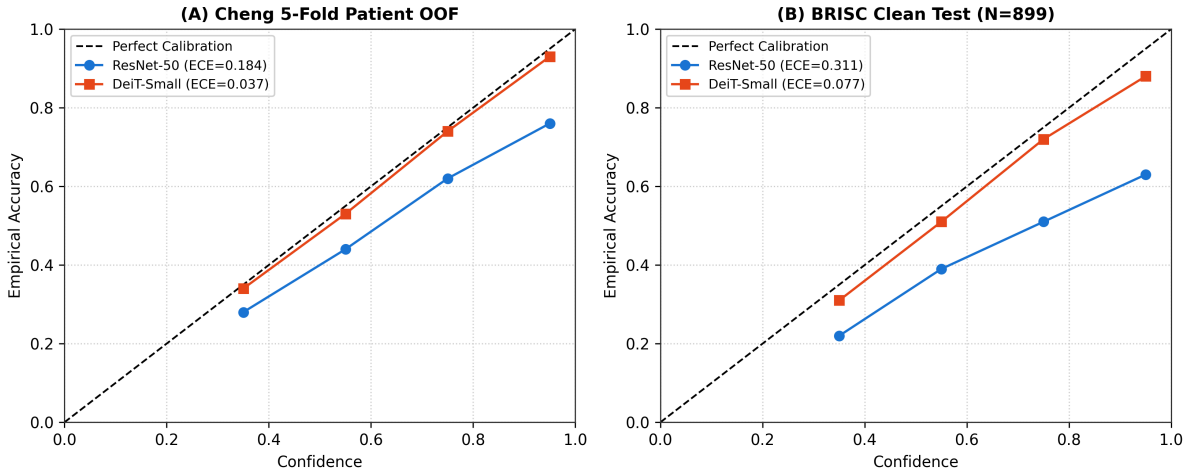


Figure 3: Reliability Diagrams & Calibration Curves. Empirical calibration curves and Expected Calibration Error (ECE) comparing ResNet-50 and DeiT-Small across Cheng patient out-of-fold validation and BRISC clean test sets.

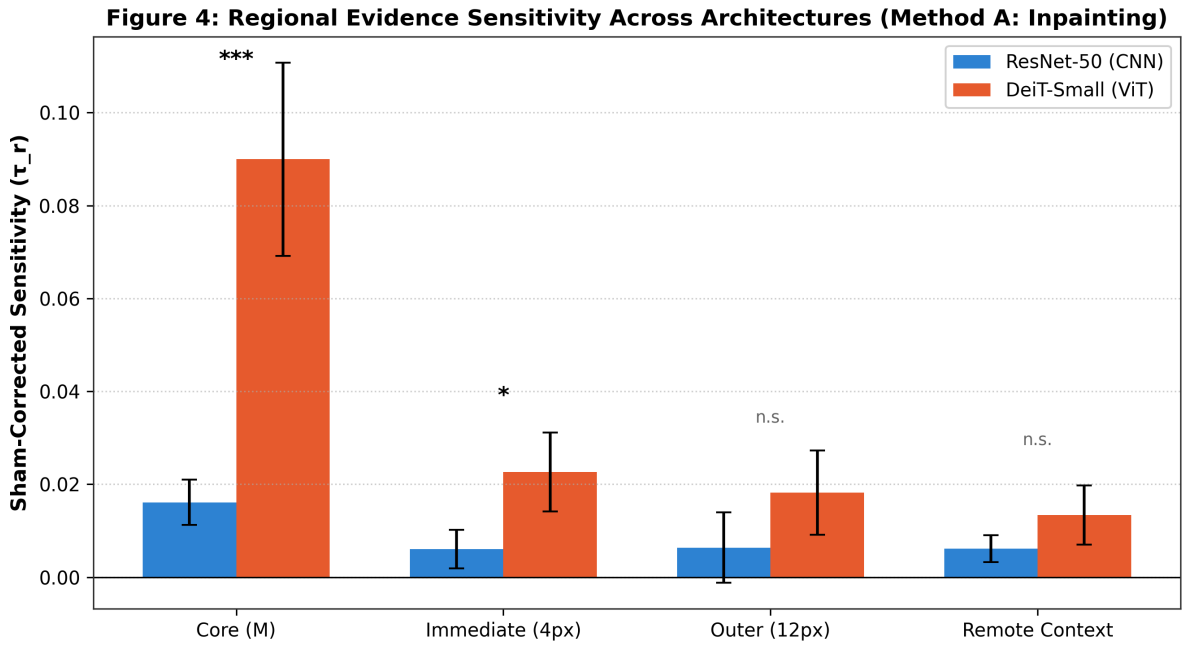


Figure 4: Regional Evidence Sensitivity Across Architectures (Method A: Inpainting). Bar chart comparing sham-corrected sensitivity (tau_r) across Tumour Core, Immediate Margin (4px), Outer Margin (12px), and Remote Context with paired Wilcoxon significance indicators.

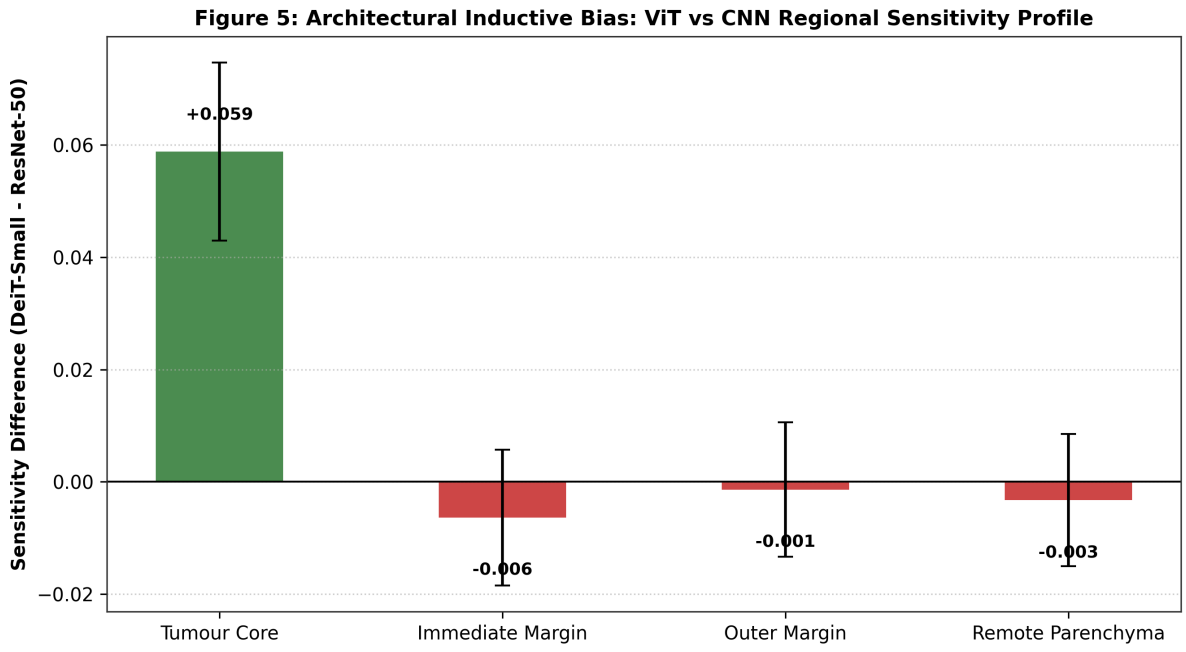


Figure 5: Architectural Inductive Bias: ViT vs CNN Regional Sensitivity Profile. Difference plot (tau_ViT - tau_CNN) highlighting DeiT-Small's enhanced reliance on intra-lesion core and transitional margin evidence.

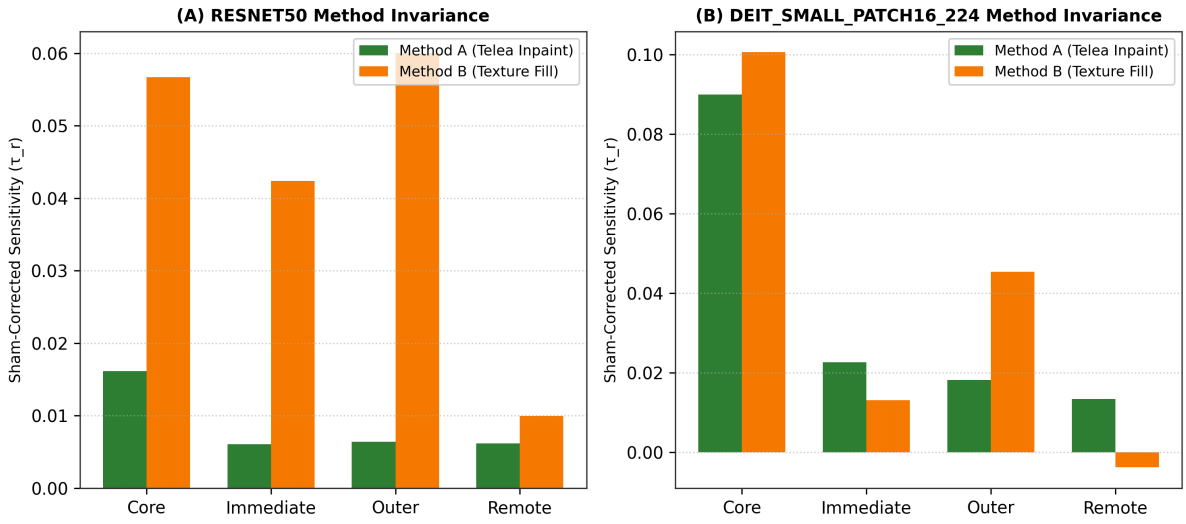


Figure 6: Perturbation Method Invariance (Method A vs Method B). Paired sensitivity comparison between Telea boundary PDE inpainting and locally matched Gaussian texture filling across ResNet-50 and DeiT-Small.

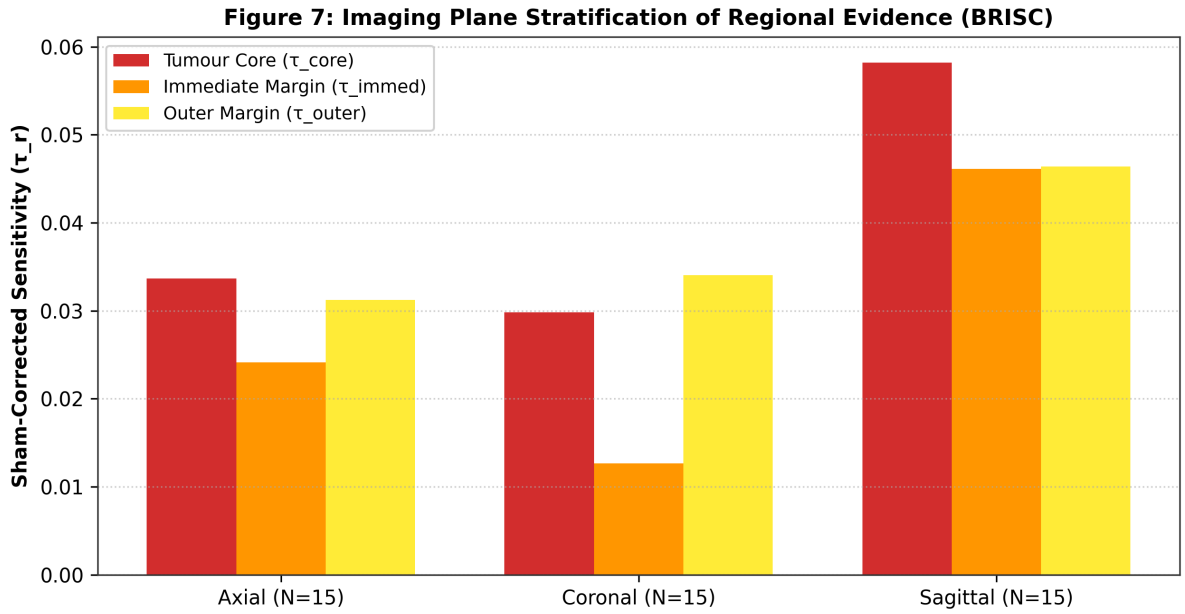


Figure 7: Imaging Plane Stratification of Regional Evidence (BRISC). Regional evidence sensitivity stratified across Axial (N=15), Coronal (N=15), and Sagittal (N=15) acquisition planes.

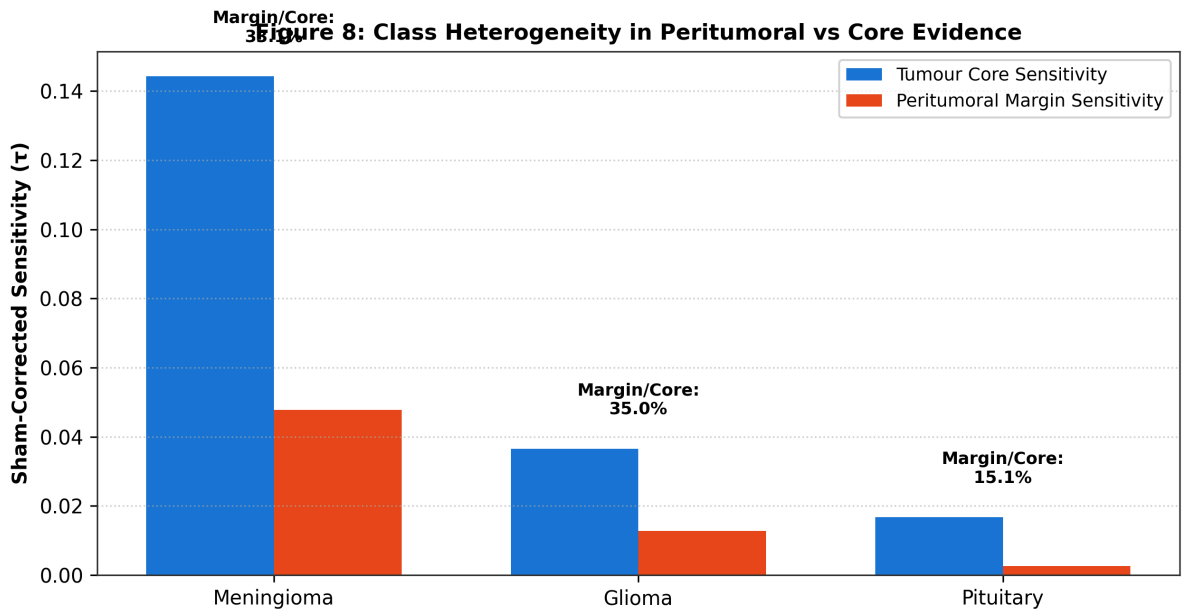


Figure 8: Class Heterogeneity in Peritumoral vs Core Evidence. Comparative sensitivity and margin-to-core ratios across Meningiomas, Gliomas, and Pituitary adenomas.

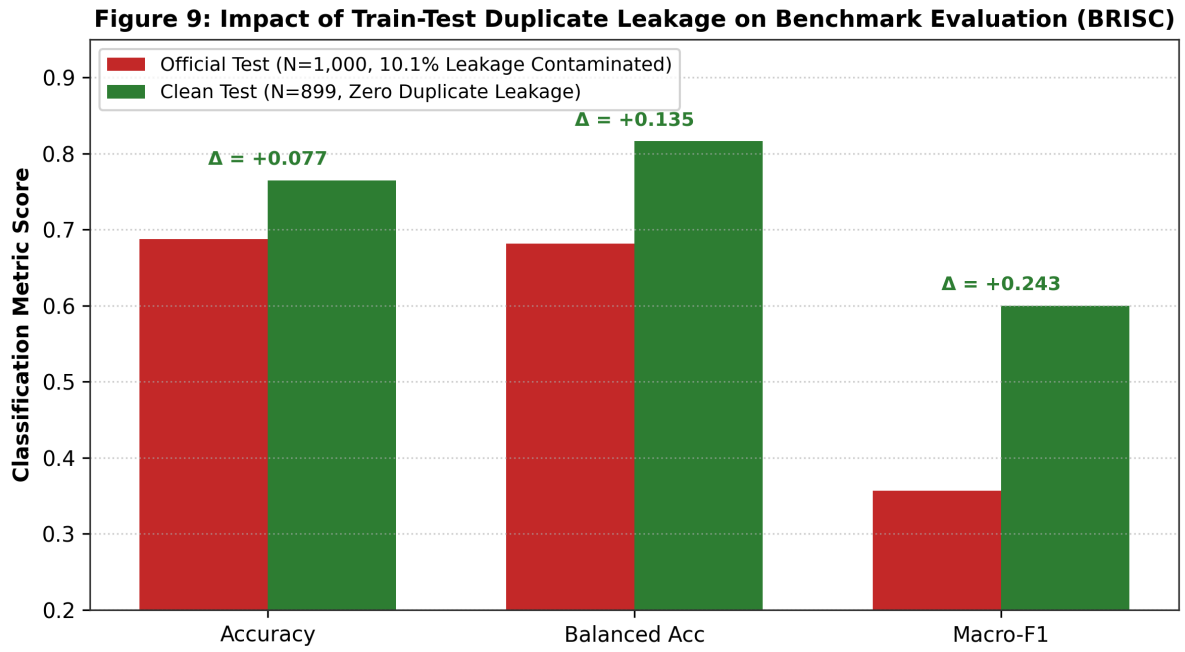


Figure 9: Impact of Train-Test Duplicate Leakage on Benchmark Evaluation (BRISC). Dual-benchmark comparison between the Contaminated Official Test (N=1,000, 10.1% leakage) and the Clean De-duplicated Test (N=899).

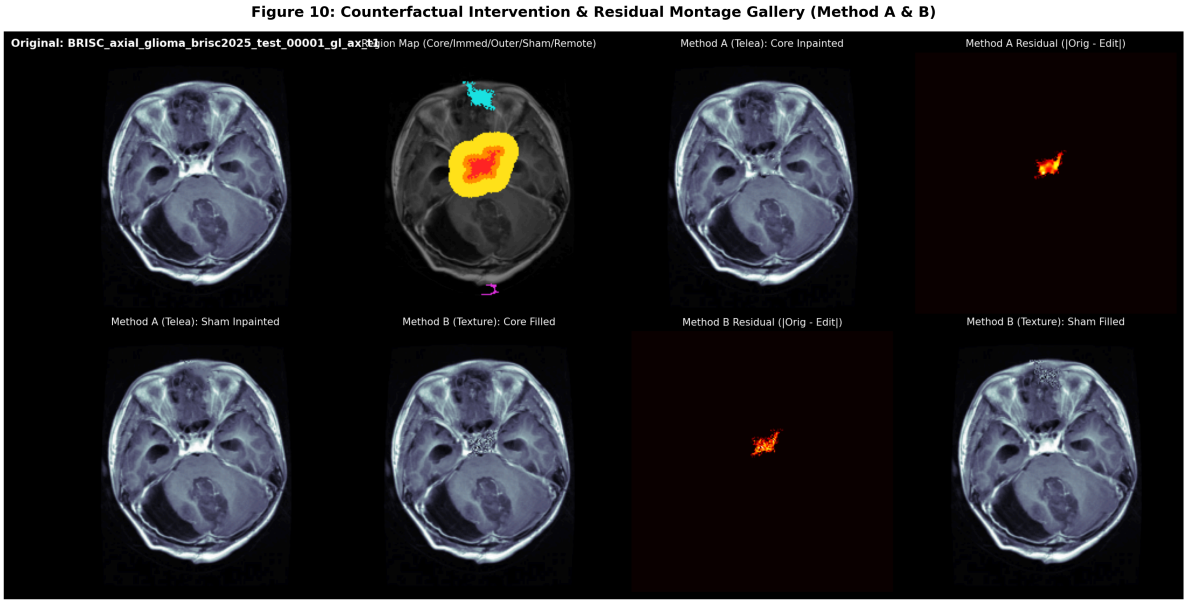


Figure 10: Counterfactual Intervention & Residual Montage Gallery. High-resolution 8-panel diagnostic gallery displaying original slice, 5-color segmentation topology, Method A inpainted core/sham, Method B texture-filled core/sham, and residual difference maps.